7.8 Acoustic Lining Of Mechanical Rooms

Mechanical room walls and ceiling of air conditioning plant room and air handling unit / fan rooms may be provided with acoustic lining of resin bonded Fibre glass with factory laminated black glass tissue as per Schedule of Quantities and as shown on the Drawings. The surface shall be cleaned and framework of 22 gage GI fabricated Channels 25 mm x 50 mm screwed back to back at 60 cm centers shall be provided vertically and horizontally so that 60 x 60 cm squares are formed. The gaps between frames shall be filled with 50 mm thick about 60 cm x 60 cm cut panels of resin bonded Fibre glass slabs. The insulation shall then be covered with 26 gauge perforated aluminum sheet, 60 cm or 120 cm wide having at least 40-50 percent perforations, fixed with sheet metal screws. Over-lapping of sheets shall be covered with aluminum beading. Acoustic lining of walls shall be terminated approximately 15 cm above the finished floor to prevent damage to insulation due to accidental water-logging in plant/AHU rooms.

7.9 Underdeck Insulation For Roof

Underdeck insulation shall be either of the following with density greater than or equal to 25 Kg/m^3 & thermal conductivity of 0.21 Btu in/ft^2 hr.º F (at 24º C as per ASTM C-518).

25mm thick Closed cell elastomeric nitrile rubber with factory finish aluminum foil protective coating.

Underdeck surface of ceiling shall be cleaned and made dirt free. Insulation panels as mentioned above shall be pasted on this surface with black CPRX compound / SR 998 rubber adhesive. 28g wire net shall be tightened around insulation so as to avoid any kind of sagging. Ends of net shall be overlapping by at least 25mm. Overlaps shall be screwed with galvanized screws to avoid rusting.

7.10 Sound / Cross talk Attenuators

Attenuators shall be installed in ducts in accordance with requirements of drawings and as included in Schedule of Quantities.

Noise levels within conditioned spaces shall be not greater than those set out in schedule below: